



2950 Niles Road, St. Joseph, MI 49085-9659, USA
269.429.0300 fax 269.429.3852 hq@asabe.org www.asabe.org

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Phosphorus Variability in the Irrigated Cropland During a growing season

***Hemendra Kumar¹, Puneet Srivastava², Jasmeet Lamba¹, Brenda V. Ortiz¹, Thomas R. Way³,
Laljeet Sangha⁴, Bijoychandra S. Takhellambam¹, Guilherme Morata¹***

¹Auburn University, ²University of Maryland, ³USDA-ARS National Soil Dynamics Laboratory Auburn, Alabama, ⁴Virginia Polytechnic Institute and State University.

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ABSTRACT. *Understanding spatial and temporal variation of soil phosphorus (P) in agricultural fields is challenging. Variation in P levels across a field can impact crop yield. The aim of this study was to investigate P variation in a cropland field during the 2019 crop growing season in the Tennessee Valley Region (TVR) of North Alabama, USA. Three different management zones (MZs) were delineated based on 10 years of historical records of crop yield, soil texture, and measured topography. Multiple locations were selected in these management zones (MZ1, MZ2, and MZ3) for soil sampling at 0-15 cm, 15-30 cm, and 30-60 cm depth at different crop growth stages. Soil samples (n = 360) at four different locations in each MZ with three replicates for each depth were collected for analysis. The precipitation data were recorded by an installed weather station in the field and irrigation records were also maintained during the season. The fertilizer application was kept uniform in each MZ of the field. The classical statistics were used to understand the spatial and temporal variation in P during the growing season. The normalized data approach was adopted to distinguish three MZs for P. The MZ1 and MZ2 zones which received an average of 1.12 times the irrigation amount of the IR3 zone, had higher P concentrations than the average P in the field for various crop growth stages. This variation in P in these MZs was found due to topographical variation, differences in soil properties, and surface runoff losses within the field. The low yield was recorded in MZ3 and high yield in MZ1 and MZ2 as compared to average yield of the field during the 2019 growing season. Therefore, it can be concluded that soil P was found to be one of the main causes that influenced the crop yield.*

Keywords. *Corn, Growing season, Irrigation, phosphorus, management zones*

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INTRODUCTION

Increasing population demands for increasing the crop yield during a growing season. An increase in the crop yield requires precision management of agricultural practices most important nutrient and water management in the crops during a growing season. The amount of irrigated land has expanded by 5% in the last 15 years in the United States (USDA-NASS, 2019). Irrigation adoption has been increasing to help farmers with improving water use efficiency, avoiding water stress, and maintaining desirable soil moisture. Therefore, adopting the best irrigation practices can also help with nutrient availability to the plants, reducing the pressure on water resource systems (Sangha et al., 2020b), and contamination to the ground and surface water during the growing season (Yost et al., 2019). Phosphorus (P) is an essential nutrient for plant growth which plays an important role in the crop yield. Excess application of P in agricultural fields can lead to contamination to surface waters by runoff or groundwater by leaching (Hanrahan et al., 2019) locally, regionally, and globally. Phosphorus in plant-available forms is also known for its deficiency that constrains the crop yield during the growing season. A numbers of factors are responsible for P loss from agricultural fields during a growing season, including fertilizer application method, rate, and timing, irrigation rate, field topography, and surface runoff. Therefore, there is always a need to adopt best management practices to minimize P losses and increase the crop yield during a growing season. Many studies have investigated spatial variation in nutrient levels in cropland, but few studies have focused on spatiotemporal variations within the same cropland during a growing season (Su et al., 2018). Therefore, more studies on variability in soil nutrients in cropland at various growth stages during a growing season are needed to help farmers to increase crop yield. The spatial and temporal variation of soil nutrients in agricultural fields has been a challenge to study around the globe (Su et al., 2018) due to laborious soil samplings in different land uses during the growing seasons and time-consuming processing of soil samples. However, more studies have been done on spatial variation in soil nutrients with less concern for temporal variability during a growing season. Also, researchers have found that temporal variability in soil nutrients was much larger than the spatial variability over the growing seasons (Santillano-Cázares et al., 2012). Various methods have been used, including geostatistical analyses, to study variability in soil nutrients in controlled conditions. However, the results from controlled experiments have a limitation of the applicability to the actual field. Various experiments have been performed to study nutrient variability from plot to watershed scales, to compare nutrient levels between plots, fields, and watersheds. However, studies on within-field nutrient variability during a growing season with implications on crop yield variability, are still scarce. Further studies are needed to identify the significance of prevailing factors influencing the spatiotemporal variability in nutrients, which are necessary for enhancing site-specific recommendations in farming practices and environmental protection and contribute to the refinement of existing decision support tools.

MATERIAL AND METHODS

STUDY AREA DESCRIPTION

This research was conducted in a farmer's field located at Town Creek of Lawrence County in the Tennessee Valley region (TVR) of North Alabama, USA (Fig. 1). The field was sown with the corn in the Spring of 2019. The plant and row spacing in the field were 15 cm and 76 cm, respectively. The field varied in elevation from 168 m to 178 m (Kumar et al., 2021). The size of the field under a center pivot irrigation system was 120 ha, and 60 ha of that total was used for this research. The slope of the field varied from 0 to 10%. The average annual precipitation during the growing season (April to October) is 759 mm with maximum and minimum precipitation of 137 mm in July and 74 mm in October, respectively (Sangha et al., 2020a). The major crop grown in the region are corn (*Zea mays* L.), cotton (*Gossypium hirsutum* L.), soybean (*Glycine max* L.), and winter wheat (*Triticum aestivum* L. Merr.) (Sangha et al., 2020b). The cornfield was seeded on March

28, 2019, and harvested on August 30, 2019. The study field was equipped with a remotely-controlled 625 m long Reinke center pivot irrigation system (Reinke Manufacturing Co., Inc., Deshler, NE, USA) with a mid-elevation spray application (MESA) sprinkler fitted with pressure regulators (Kumar et al., 2021). The study field was delineated in three different irrigation management zones (MZs) to set up the variable rate irrigation technology to schedule different irrigation prescriptions during the growing season. The field was delineated based on soil texture, topography, 10 years of historical yield data, farmers' experience, and visual conditions of the field.

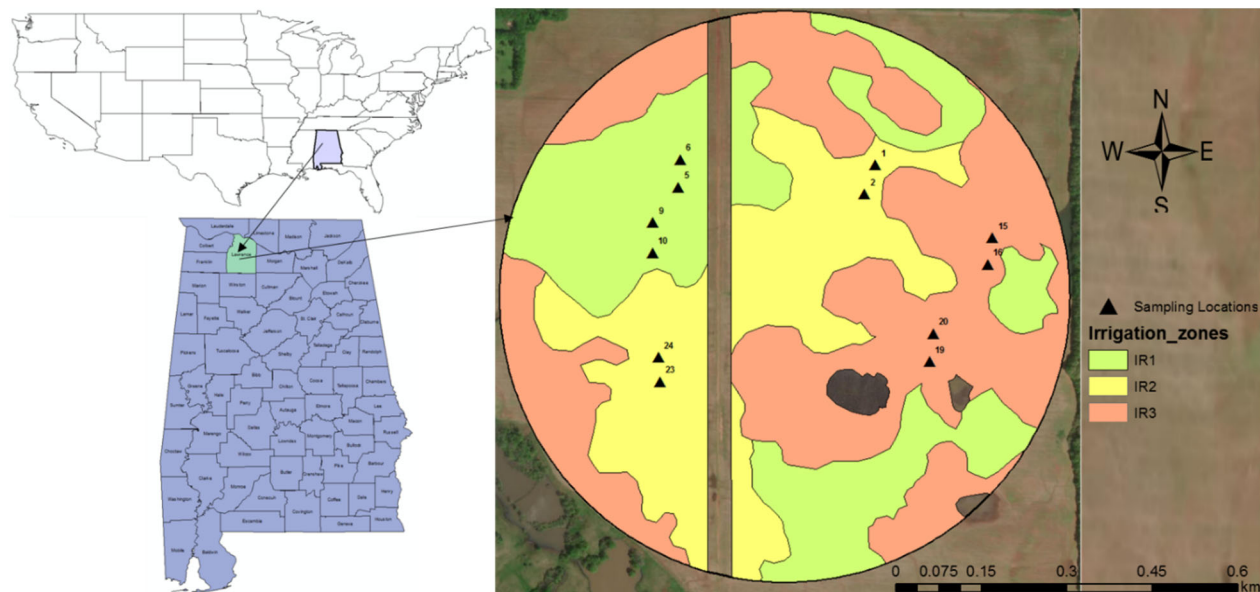


Figure 1. Study field with soil sampling locations for phosphorus analysis

SAMPLE COLLECTION, PROCESSING, AND DATA ANALYSIS

Multiple locations were selected in these management zones (IR1, IR2, and IR3) for soil sampling at 0-15 cm, 15-30 cm, and 30-60 cm depths at different crop growth stages (five times) during the growing season (Fig. 1). A total of 540 (3 depths x 3 replicates x 12 locations x 5 sampling times) soil samples at four different locations in each IR zone with three replicates for each depth were collected for analysis from before sowing to after harvesting the corn. The soil samples were collected using hand operated soil auger (AMS, Inc., American Falls, ID, USA). Three replicates of collected soil samples were compositely mixed for each depth at each location. The soil samples were stored in a refrigerator at 4°C until analyzed. The samples were analyzed for total P, ortho-phosphorus (OP), and soil organic matter (SOM). Poultry litter samples were also stored at 4 °C until analyzed. The soil and poultry litter samples were oven-dried at 55 °C before grinding. The oven-dried soil samples were ground and passed through a 2-mm sieve for chemical analysis. The soil samples were ground using a soil grinder. The grinder was cleaned with a vacuum cleaner between the samples. Poultry litter samples were ground using a pestle and mortar and passed through a 2-mm sieve before analysis. An automated method based on colorimetry was adopted for OP determination in the samples. The OP ions react with ammonium molybdate and antimony potassium tartrate under acidic conditions to form a molybdate-phosphate complex. Total P in soil and litter samples was determined by acid digestion using persulfate Kjeldahl digesting solution (American Public Health Association, 1998). All the standard solutions including blanks were also digested to keep a similar matrix for the analysis. A 0.5 g sample was digested in a 6 ml digesting solution and centrifuged before the analysis. A Lachat automated flow injection analyzer (Quikchem series 2) (Hach Company, Loveland, CO, USA) was used to analyze both total P and OP in the soil and litter samples. All the statistical analysis was performed at 0.05 significant level in this study.

The classical statistics were used to understand the spatial and temporal variation in total P during the growing season.

The depth weighted average of total P and OP was computed at each sampling location to understand the overall variation in P in the soil layers in the field. The total P and OP data were normalized based on the average concentration to understand the variation in total P or OP at different sampling locations at different sampling times in the cropland during the growing season. The relative difference was estimated for each location and each sampling time.

RESULTS AND DISCUSSION

PHOSPHORUS VARIABILITY DURING THE GROWING SEASON

The weighted average of total P concentrations in three soil layers (0-15 cm, 15-30 cm, and 30-60 cm) at each location, to understand the variation in the entire soil profile, were computed for each sampling stage. The sampling locations in the management zones MZ1 (5, 6, 9, 10) and MZ2 (1, 2, 23, 24) indicated higher total P concentrations than average total P in the field during growing season. However, the locations in the MZ3 (15, 16, 19, and 20) had lower total P concentrations than the average total P in the field during growing season (Fig. 2). The application of the fertilizers and poultry litter was kept uniform over the entire field. However, variation in the field topography, soil physical and chemical properties, and field hydrology (Plach et al., 2019) might resulted in the variable soil P levels. Within-field variability was also observed in the OP levels. The OP variation followed a pattern similar to that for total P variation in the respective irrigation zones during the crop growing period. The concentration of P affects plant reproduction, plant stalk strength, plant growth, and crop yield (Beegle, 2002). During the growing season, it was noticed that the plants in the MZ3 zone were smaller in height than in MZ1 and MZ2 (Fig. 3), which ultimately delayed the maturity of the crop in the zone. The topographical variations can be understood using the slope variations in the field (Fig. 4). A high slope area increases the chance of lateral water movement and reduces the infiltration in deeper soil layers (Huat et al., 2006). The locations in the zones MZ1 (5, 6, 9, 10) and MZ2 (1, 2, 23, 24) were at the flatter slope in the field. However, the locations in the MZ3 (15, 16, 19, 20) were at a steep slope in the field. It might be possible that the runoff generation was one of the dominant factors to lower the P in MZ3 zone.

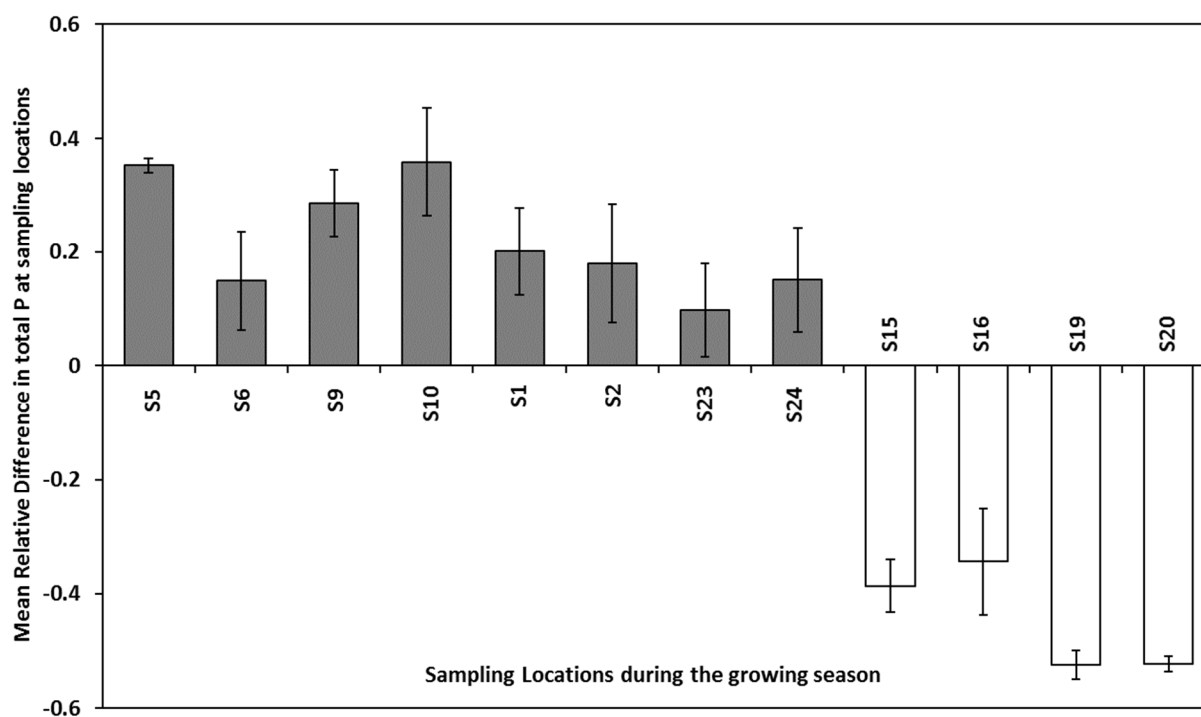


Figure 2. Seasonal total P distribution during the growing season. Each half bar represents one standard deviation.

The relative difference was calculated to understand the variability and to identify the high or low P zones as compared

to the average P at each sampling stage during the growing season in the field. Zones IR1 (5, 6, 9, and 10) and IR2 (1, 2, 23, and 24) had relatively high total P; however, IR3 (15, 16, 19, and 20) was total P deficient relative to the average total P within the field at each sampling stage during the growing season. The surface runoff and erodibility due to a higher slope in the MZ3 zone might induce P loss during the growing season. According to the soil hydraulic properties (soil water retention curves were developed using the HYPROP 2 instrument), the saturated hydraulic conductivity for the MZ3 soil (1.49 cm/day) was lower as compared to MZ1 (12.6 cm/day) and MZ2 (10.6 cm/day). Therefore, the higher slope and lower hydraulic conductivity in the MZ3 zone could have increased surface runoff, thereby causing the P loss during the growing season. When the rainfall intensity is greater than the saturated hydraulic conductivity of the soils, surface runoff occurs along the slope (Huat et al., 2016). Based on water input data, the entire field received the same precipitation events and MZ3 received the least irrigation amount during the growing season. The Kruskal-Wallis test explained the non-significant variations in the spatiotemporal mean of total P, in the 0-15 cm soil layer, between the MZ1 and MZ2 during the growing season. However, significant variations in the spatiotemporal mean of total P were explained between MZ1 and MZ3, and MZ2 and MZ3 in the 0-15 cm soil depth during the growing season. A similar pattern in total P variation was observed in the 15-30 cm soil layer among all three irrigations zones. However, a significant total P variation at the deepest soil layer (30-60 cm) was found between the MZ1 and MZ2, MZ1 and MZ3, and MZ2 and MZ3 irrigation zones during the entire growing season. The MZ1 zone was functioning as a valley shape for P with fast water movement through the soil layers and IR3 as a ridge shape for P with slow water movement (Fig. 4). The valley shape of the MZ1 zone may have caused development of a P pool (lower slope zone), which promoted leaching (Fig. 4). However, due to the ridge behavior of the MZ3 zone, the surface runoff was dominant during the growing season. The status of the IR2 zone was resting between the states of MZ1 and MZ3. The MZ2 zone had properties including topography, slope, and hydraulic conductivity between those of the MZ1 and MZ3 zones.



Figure 3. Plants in (a) MZ1 and MZ2, and (b) MZ3.

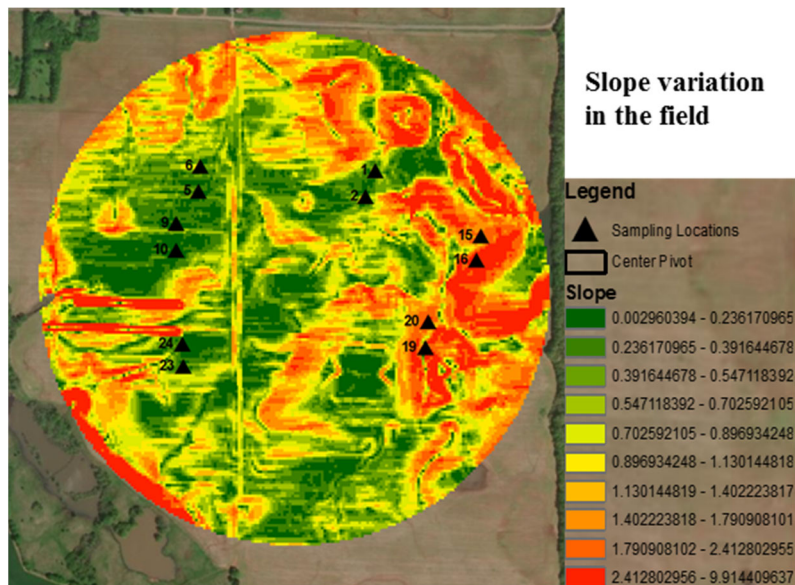


Figure 4. Slope variability in the cornfield.

CROP YIELD VARIABILITY

The yield produced in the MZ zones varied from 1569 kg/ha to 18831 kg/ha. This corn grain yield is adjusted to a 15.5 % moisture. A significant difference was recorded in the crop yield in IR1 and IR3 zones of the same field. The least yield was produced in the IR3 zone. The least yield was produced in the MZ3 zone. However, a higher yield was produced in the other two irrigation zones of the field. All the irrigation zones received the same number of irrigation (with a minimum amount of irrigation in IR3) and precipitation events during the growing season. During the soil sampling at different times in the season, we noticed that the plants were smaller and were more stressed in the MZ3 zone as compared to MZ1 and MZ2. Less availability of P due to loss can be one of the main reasons for the crop yield variability in the irrigation zones. We also calculated Spearman's correlation coefficient between yield and temporal total P. It showed a positive significant correlation of 0.68 ($p < 0.01$), which indicates that increasing P can increase crop yield. Based on these results, it can be correlated that P loss along with some other factors can be one of the main reasons for the crop yield variability in the irrigation zones.

SUMMARY AND CONCLUSIONS

We have presented preliminary results from the cornfield study performed in the TVR in North Alabama, USA. The results showed significant variations in P and crop yield within the field. Our assessment of P variation in the MZ zones within the same field showed that MZ3 had lower total P and OP than MZ1 and MZ2 during the growing season. Consequently, a similar pattern in crop yield was also shown in the cornfield, which could be a result of P variability during the growing season. Consistent management practices (irrigation, fertilizer, and manure) were applied across the field; however, different terrain attributes affected P loss and crop yield within the IRs of the same field. Therefore, it is of utmost importance to study the temporal variability in nutrients in different parts of a crop field to increase crop yield, reduce loss of nutrients, and reduce negative environmental aspects.

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